

ProFACT

Protein FASTA Analysis & Comparison Tool

*A Python CLI tool for protein FASTA
quality control, duplicate detection and
dataset comparison*



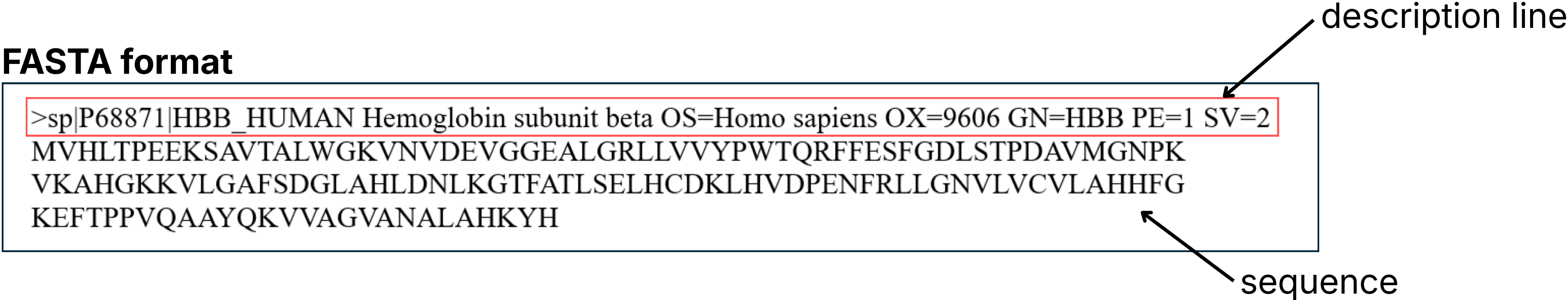
```
$ profact --help  
> stats | validate  
> duplicates | compare  
> report
```

Developed by
Wojciech Laskowski
Wojciech Moryl
Karolina Winczewska

Protein FASTA Files

Proteins are large biomolecules made of amino acids, stored in FASTA text format.

FASTA format



The diagram illustrates the FASTA format with a sample entry for Hemoglobin subunit beta. The entry is enclosed in a black rectangular box. The first line, which is the description line, is highlighted with a red border and is labeled with an arrow pointing to it from the text 'description line' located to the right of the box. The subsequent three lines, which contain the amino acid sequence, are labeled with an arrow pointing to them from the text 'sequence' located below and to the right of the box.

```
>sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Homo sapiens OX=9606 GN=HBB PE=1 SV=2
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
VKAHGKKVLGAFSDGLAHLNDNLKGTfATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
KEFTPPVQAAYQKVVAGVANALAHKYH
```

Common sources: UniProt, RefSeq, and Ensembl/NCBI.

Why ProFACT?

Protein FASTA files often need quality checks before further analysis.

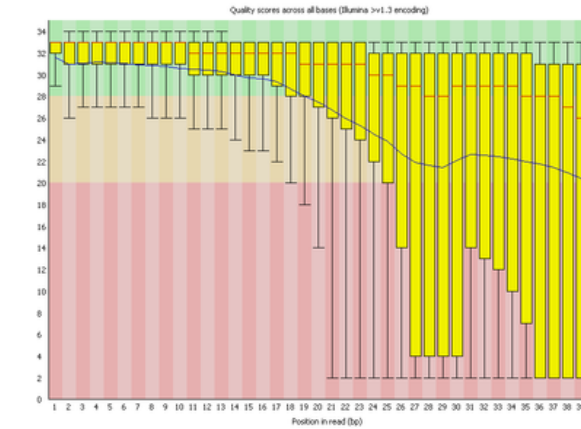
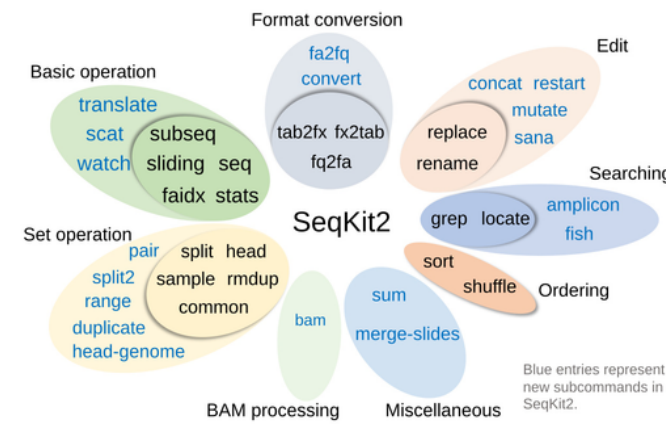
Typical QC questions

- Are all sequences valid?
- Are there unknown residues?
- Are there invalid amino acid characters?
- Are there duplicate IDs?
- What changed between dataset versions?

Workflow challenge

These checks often require separate tools or custom scripts, leading to fragmented workflows.

Similar Tools



SeqKit

Fast sequence toolkit; supports filtering, statistics and duplicate detection.

FastQC

Quality control for sequencing reads, mainly nucleotide data.

protfasta

Protein FASTA filtering and validation utilities that filter sequences by length and remove entries containing invalid amino acids.

BioPython

Large general-purpose bioinformatics library.

ProFACT does not replace these tools. Its value is integration: common protein FASTA QC tasks in one small CLI.

ProFACT Functionality

Inspect

`profact stats`

Validate

`profact validate`

Find duplicates

`profact duplicates`

Compare

`profact compare`

Report

`profact report`

One package, five commands, consistent output formats.



ProFACT

Protein FASTA Analysis and Comparison Tool

Validation

Command
`profact validate`

Exit Code 1

Returned when invalid records are found.

Detection Criteria

Errors

- Empty sequences
- Invalid amino acid characters

Warnings

- Unknown residue X
- Stop symbol *
- Non-standard amino acids (B, Z, J)

Validation

Metric	Value
Total Records	4403
Valid Records	4400
Invalid Records	3
Records With X	7
Records With Stop	0
Empty Records	0

Duplicate Detection

Command
profact duplicates

Detects

- Repeated sequence IDs
- Identical sequences under different IDs
- Exact duplicate sequence clusters

Example Cluster Output

Cluster 1
length: 35
records: 2
IDs: LDRA_ECOLI, LDRC_ECOLI

Duplicates

Metric	Value
Total Records	4403
Duplicate Id Count	0
Identical Sequence Cluster Count	13
Records In Identical Sequence Clusters	54

No duplicate IDs found.

Important Note

Clustering uses exact sequence matching, not similarity search.

Comparing FASTA Files

Command

```
profact compare -f1 old.fasta -f2  
new.fasta
```

Exit Code 1

Returned when differences are found.

Identified Differences

- Added IDs
- Removed IDs
- Changed sequences
- Changed sequence lengths
- Changed duplicate clusters

Output Formats



Sequence Statistics

Command

`profact stats`

Computed Metrics

- sequence count
- total sequence length
- minimum and maximum length
- mean and median length
- N50
- amino acid composition

Statistics report

Protein FASTA summary statistics and amino acid composition.

SEQUENCES

4403

TOTAL LENGTH

1354442

N50

393

MEDIAN LENGTH

271

Summary statistics

Metric	Value
Sequence Count	4403
Total Length	1354442
Min Length	8
Max Length	2339
Mean Length	307.6179877356348
Median Length	271
N50	393



TXT



TSV



JSON



HTML

Reports

Command

```
profact report -i proteins.fasta -fmt html  
-o output/report.html
```

HTML Report Includes

- Summary cards
- Detailed statistics tables
- Amino acid composition chart
- Validation summary
- Duplicate cluster table

ProFACT full report

File: data/uniprotkb_proteome_UP000000625_2026_06_06.fasta

SEQUENCES

4403

TOTAL LENGTH

1354442

N50

393

VALIDATION

**Issues
found**

DUPLICATE
CLUSTERS

13

Output Formats



TXT



TSV



JSON



HTML

Purpose: Readable overview of FASTA quality, statistics and duplicates.

Reproducibility

Reproducible Example

```
bash scripts/run_examples.sh
```

33 Automated Tests Passed

What it does

- Runs automated tests to ensure software integrity
- Analyzes included FASTA datasets for protein sequence checks
- Generates comprehensive statistics, validation, and duplicate reports
- Compares two sample FASTA files to identify structural differences
- Writes all processed outputs directly to the **output/**directory

Why it matters: The project can be checked and reproduced with one simple command.

Summary

What we built

ProFACT is a Python CLI tool for protein FASTA quality control and dataset comparison.

Main Features

- Sequence statistics
- Validation of protein FASTA records
- Duplicate ID & sequence detection
- Comparison of two FASTA datasets
- Text, TSV, JSON and HTML reports

Project Quality

- Sample FASTA datasets
- Reproducible script
- 30 Automated tests passed

Final message: ProFACT combines common protein FASTA checks into one small, testable and easy-to-run tool.